

md2tex - Markdown to LaTeX Converter

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md2tex

md2tex is a powerful, customizable open-source converter that safely translates your standard and GitHub Flavored Markdown (GFM) logic into pure TeX syntax.

Originally built as a simple parser, md2tex has been **completely modernized** into a robust parsing pipeline powered by an Intermediate Representation (IR) AST structure that guarantees 100% stable parsing without colliding formatting conflicts.

Features

- **Strict AST Architecture:** Guaranteed reliable markdown block parsing before inline TeX escaping prevents artifacts.
- **TUI Dashboard:** Native Textual Terminal User Interface for configuration-free execution.
- **Full Template Control:** Automatically wrap logic in predefined headers, or inject converted contents seamlessly into custom TeX template `@@BODYTOKEN@@` blocks.
- **GFM Support:** Full implicit translation of Tables, Strikethroughs, Code Blocks, Headers, References from standard Markdowns directly into `tabular`, `sout`, `minted`, `section`, etc.
- **Quotes:** Native parameter translation into French inline formatting `\{}enquote{}` via `csquotes`.

Installation

It is highly recommended to manage implementations via modern package tooling uv.

1. Simple installation

```
1 git clone git@github.com:Malk97sc/md2tex.git
2 cd md2tex
3
4 # We strongly recommend using the interactive TUI
5 pip install ".[tui]"
```

2. Developing

```
1 git clone git@github.com:Malk97sc/md2tex.git
2 cd md2tex
3
4 # Setup tests and developer linting suites
5 uv sync
```

3. LaTeX Requirements

For local PDF compilation of your `.tex` files, you need `pdflatex` installed. On Unix systems, you can install TeX Live:

Ubuntu / Debian / Mint:

```
1 sudo apt update
2 sudo apt install texlive-latex-extra texlive-fonts-recommended
```

Fedora / RHEL:

```
1 sudo dnf install texlive-scheme-medium
```

macOS (via Homebrew):

```
1 brew install --cask mactex-no-gui
```

Usage

Place your documents

To use `md2tex`, we recommend placing your `.md` files inside the `myfiles` folder.

Terminal Interface (TUI) Mode - Recommended

Instead of memorizing flags, directly launch a beautiful TUI configuration dashboard:

```
1 md2tex tui
2
3 # Or
4 md2tex interactive
```

The TUI can now convert to `.tex` and optionally compile to `.pdf` with `pdflatex` (2 passes), including an explicit `shell-escape` toggle.

Direct CLI Conversion

If you're looking for headless automations or quick CLI overrides:

```
1 # General convert (Defaults Article class output to output/<filename>.
   tex)
2 md2tex myfiles/document.md
3 # Or
4 md2tex path/to/document.md
5
6 # Build full encapsulated '.tex' compiling with headers
7 md2tex path/to/document.md -c
8
9 # Advanced custom translation output
10 md2tex source.md -c -t ./utils/template.tex -o ./export/export.tex -d
   book -u -f
```

Documentation

For a complete understanding of how this tool abstracts parsing structures, as well as TUI capabilities, please review the documentation directory:

- System Architecture Overview
- Terminal User Interface (TUI)
- CLI and TUI Parameters (Step by Step)

License and Authors

Developed by **Paul Kervegan** in August 2022. Modernized, refactored, and updated with TUI integration by me, *Malak* in 2026.

Released Open Source under GNU GPL v3.